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Question 1

Employee.java

```
package com.workshop.employee.model;

public class Employee {
    private int employeeId;
    private String name;
    private String department;
    private double salary;

    public int getEmployeeId() {
        return employeeId;
    }

    public void setEmployeeId(int employeeId) {
        this.employeeId = employeeId;
    }

    public String getName() {
        return name;
    }

    public void setName(String name) {
        this.name = name;
    }

    public String getDepartment() {
        return department;
    }

    public void setDepartment(String department) {
        this.department = department;
    }

    public double getSalary() {
        return salary;
    }

    public void setSalary(double salary) {
        this.salary = salary;
    }
}
```

EmployeeDAO.java

```
package com.workshop.employee.dao;

import org.springframework.dao.EmptyResultDataAccessException;
import org.springframework.jdbc.core.JdbcTemplate;

import com.workshop.employee.model.Employee;

public class EmployeeDAO {
    private final JdbcTemplate jdbcTemplate;

    public EmployeeDAO(JdbcTemplate jdbcTemplate) {
        this.jdbcTemplate = jdbcTemplate;
    }

    public Employee findEmployeeById(int employeeId) {
        String sql = "SELECT employee_id, name, department, salary FROM employee WHERE employee_id = ?";

        try {
            return jdbcTemplate.queryForObject(sql, (resultSet, rowNum) -> {
                Employee employee = new Employee();
                employee.setEmployeeId(resultSet.getInt("employee_id"));
                employee.setName(resultSet.getString("name"));
                employee.setDepartment(resultSet.getString("department"));
                employee.setSalary(resultSet.getDouble("salary"));
                return employee;
            }, employeeId);
        } catch (EmptyResultDataAccessException exception) {
            return null;
        }
    }
}
```

EmployeeController.java

```
package com.workshop.employee.controller;

import org.springframework.stereotype.Controller;
import org.springframework.ui.Model;
import org.springframework.web.bind.annotation.GetMapping;
import org.springframework.web.bind.annotation.PostMapping;
import org.springframework.web.bind.annotation.RequestParam;

import com.workshop.employee.dao.EmployeeDAO;
import com.workshop.employee.model.Employee;

@Controller
public class EmployeeController {
    private final EmployeeDAO employeeDAO;

    public EmployeeController(EmployeeDAO employeeDAO) {
        this.employeeDAO = employeeDAO;
    }

    @GetMapping("/")
```

```

    public String showSearchPage() {
        return "search";
    }

    @PostMapping("/search")
    public String searchEmployee(@RequestParam("employeeId") int employeeId, Model
model) {
        Employee employee = employeeDAO.findEmployeeById(employeeId);

        if (employee == null) {
            model.addAttribute("message", "Employee not found.");
        } else {
            model.addAttribute("employee", employee);
        }

        return "employee";
    }
}

```

[search.jsp](#)

```

<%@ page contentType="text/html; charset=UTF-8" language="java" %>
<!DOCTYPE html>
<html>
<head>
    <title>Employee Search Portal</title>
</head>
<body>
    <h2>Employee Search Portal</h2>

    <form action="search" method="post">
        <label>Employee ID:</label>
        <input type="number" name="employeeId" required>
        <br><br>

        <input type="submit" value="Search">
    </form>
</body>
</html>

```

[employee.jsp](#)

```

<%@ page import="com.workshop.employee.model.Employee" %>
<%@ page contentType="text/html; charset=UTF-8" language="java" %>
<%
    Employee employee = (Employee) request.getAttribute("employee");
    String message = (String) request.getAttribute("message");
%>
<!DOCTYPE html>
<html>
<head>
    <title>Employee Details</title>
</head>
<body>
    <h2>Employee Details</h2>

```

```

<% if (message != null) { %>
    <p><%= message %></p>
<% } %>

<% if (employee != null) { %>
    <p>Employee ID: <%= employee.getEmployeeId() %></p>
    <p>Name: <%= employee.getName() %></p>
    <p>Department: <%= employee.getDepartment() %></p>
    <p>Salary: <%= employee.getSalary() %></p>
<% } %>

    <p><a href="{pageContext.request.contextPath}/">Search another
employee</a></p>
</body>
</html>

```

[dispatcher-servlet.xml](#)

```

<?xml version="1.0" encoding="UTF-8"?>
<beans xmlns="http://www.springframework.org/schema/beans"
    xmlns:context="http://www.springframework.org/schema/context"
    xmlns:mvc="http://www.springframework.org/schema/mvc"
    xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
    xsi:schemaLocation="
        http://www.springframework.org/schema/beans
https://www.springframework.org/schema/beans/spring-beans.xsd
        http://www.springframework.org/schema/context
https://www.springframework.org/schema/context/spring-context.xsd
        http://www.springframework.org/schema/mvc
https://www.springframework.org/schema/mvc/spring-mvc.xsd">

    <context:component-scan base-package="com.workshop.employee" />
    <mvc:annotation-driven />

    <bean id="dataSource"
class="org.springframework.jdbc.datasource.DriverManagerDataSource">
        <property name="driverClassName" value="com.mysql.cj.jdbc.Driver" />
        <property name="url" value="jdbc:mysql://localhost:3306/NIET" />
        <property name="username" value="root" />
        <property name="password" value="" />
    </bean>

    <bean id="jdbcTemplate" class="org.springframework.jdbc.core.JdbcTemplate">
        <property name="dataSource" ref="dataSource" />
    </bean>

    <bean id="employeeDAO" class="com.workshop.employee.dao.EmployeeDAO">
        <constructor-arg ref="jdbcTemplate" />
    </bean>

    <bean
class="org.springframework.web.servlet.view.InternalResourceViewResolver">
        <property name="prefix" value="/WEB-INF/views/" />
        <property name="suffix" value=".jsp" />
    </bean>
</beans>

```

web.xml

```
<?xml version="1.0" encoding="UTF-8"?>
<web-app xmlns="http://xmlns.jcp.org/xml/ns/javaee"
          xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
          xsi:schemaLocation="http://xmlns.jcp.org/xml/ns/javaee
http://xmlns.jcp.org/xml/ns/javaee/web-app_3_1.xsd"
          version="3.1">

    <servlet>
        <servlet-name>dispatcher</servlet-name>
        <servlet-class>org.springframework.web.servlet.DispatcherServlet</servlet-
class>
        <load-on-startup>1</load-on-startup>
    </servlet>

    <servlet-mapping>
        <servlet-name>dispatcher</servlet-name>
        <url-pattern>/</url-pattern>
    </servlet-mapping>
</web-app>
```

database.sql

```
CREATE DATABASE IF NOT EXISTS NIET;
```

```
USE NIET;
```

```
CREATE TABLE IF NOT EXISTS employee (
    employee_id INT PRIMARY KEY,
    name VARCHAR(50),
    department VARCHAR(50),
    salary DOUBLE
);
```

```
INSERT INTO employee (employee_id, name, department, salary) VALUES
(101, 'Aarav Sharma', 'Computer Science', 45000),
(102, 'Ananya Gupta', 'Information Technology', 48000),
(103, 'Rohan Verma', 'Accounts', 38000),
(104, 'Priya Singh', 'Human Resources', 42000),
(105, 'Kunal Mehta', 'Sales', 40000)
ON DUPLICATE KEY UPDATE
name = VALUES(name),
department = VALUES(department),
salary = VALUES(salary);
```

[Output](#)

[Landing Page](#)

Employee Search Portal

Employee ID:

[Search page](#)

Employee Details

Employee ID: 101

Name: Aarav Sharma

Department: Computer Science

Salary: 45000.0

[Search another employee](#)

Question 2

[pom.xml](#)

```
<project xmlns="http://maven.apache.org/POM/4.0.0"
    xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
    xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
https://maven.apache.org/xsd/maven-4.0.0.xsd">
    <modelVersion>4.0.0</modelVersion>

    <groupId>com.workshop</groupId>
    <artifactId>printer-setter-injection</artifactId>
    <version>1.0.0</version>

    <properties>
        <maven.compiler.release>17</maven.compiler.release>
        <project.build.sourceEncoding>UTF-8</project.build.sourceEncoding>
    </properties>

    <dependencies>
        <dependency>
            <groupId>org.springframework</groupId>
            <artifactId>spring-context</artifactId>
            <version>6.1.14</version>
        </dependency>
    </dependencies>

    <build>
        <plugins>
            <plugin>
                <groupId>org.apache.maven.plugins</groupId>
                <artifactId>maven-compiler-plugin</artifactId>
                <version>3.13.0</version>
            </plugin>
            <plugin>
                <groupId>org.codehaus.mojo</groupId>
                <artifactId>exec-maven-plugin</artifactId>
                <version>3.5.0</version>
                <configuration>
                    <mainClass>com.workshop.printer.App</mainClass>
                </configuration>
            </plugin>
        </plugins>
    </build>
</project>
```

[Ink.java](#)

```
package com.workshop.printer.model;

public class Ink {
    private String brand;
    private String color;
    private String type;

    public String getBrand() {
```

```

        return brand;
    }

    public void setBrand(String brand) {
        this.brand = brand;
    }

    public String getColor() {
        return color;
    }

    public void setColor(String color) {
        this.color = color;
    }

    public String getType() {
        return type;
    }

    public void setType(String type) {
        this.type = type;
    }
}

```

[Printer.java](#)

```

package com.workshop.printer.model;

public class Printer {
    private String printerName;
    private String printerType;
    private Ink ink;

    public String getPrinterName() {
        return printerName;
    }

    public void setPrinterName(String printerName) {
        this.printerName = printerName;
    }

    public String getPrinterType() {
        return printerType;
    }

    public void setPrinterType(String printerType) {
        this.printerType = printerType;
    }

    public Ink getInk() {
        return ink;
    }

    public void setInk(Ink ink) {
        this.ink = ink;
    }

    public void displayPrinterDetails() {

```



```

        System.out.println("Printer Details");
        System.out.println("Printer Name: " + printerName);
        System.out.println("Printer Type: " + printerType);
        System.out.println("Ink Brand: " + ink.getBrand());
        System.out.println("Ink Color: " + ink.getColor());
        System.out.println("Ink Type: " + ink.getType());
    }
}

```

[AppConfig.java](#)

```

package com.workshop.printer.config;

import org.springframework.context.annotation.Bean;
import org.springframework.context.annotation.Configuration;

import com.workshop.printer.model.Ink;
import com.workshop.printer.model.Printer;

@Configuration
public class AppConfig {
    @Bean
    public Ink ink() {
        Ink ink = new Ink();
        ink.setBrand("HP Original Ink");
        ink.setColor("Black");
        ink.setType("Cartridge");
        return ink;
    }

    @Bean
    public Printer printer(Ink ink) {
        Printer printer = new Printer();
        printer.setPrinterName("OfficeJet 250");
        printer.setPrinterType("Inkjet Printer");
        printer.setInk(ink);
        return printer;
    }
}

```

[App.java](#)

```

package com.workshop.printer;

import org.springframework.context.annotation.AnnotationConfigApplicationContext;

import com.workshop.printer.config.AppConfig;
import com.workshop.printer.model.Printer;

public class App {
    public static void main(String[] args) {
        try (AnnotationConfigApplicationContext context = new
AnnotationConfigApplicationContext(AppConfig.class)) {
            Printer printer = context.getBean(Printer.class);

```

```

        printer.displayPrinterDetails();
    }
}

```

Output

```

[neo] Scanning for projects...
[neo] -----< com.workshop.printer.setter.injection >-----
[neo] building printer.setter.injection 1.0.0
[neo]   from pom.xml
[neo] -----[ jar ]-----
[neo] --- clean:3.2.0:clean (default-clean) @ printer.setter.injection ---
[neo] Deleting E:\Programming\college_workshop\3rd_Year\SEM_V\Advance_Java_workshop\day_7\Question_3\PrinterSetterInjection\target
[neo] --- resources:3.3.0:resources (default-resources) @ printer.setter.injection ---
[neo] skip non existing resourceDirectory E:\Programming\college_workshop\3rd_Year\SEM_V\Advance_Java_workshop\day_7\Question_3\PrinterSetterInjection\src\main\resources
[neo] --- compiler:3.13.0:compile (default-compile) @ printer.setter.injection ---
[neo] Recompiling the module because of changed source code.
[neo] Compiling 4 source files with javac [debug release 17] to target\classes
[neo] --- exec:3.0.0:java (default-cli) @ printer.setter.injection ---
Printer Details
Printer Name: Officejet 250
Printer Type: Inkjet Printer
Ink Brand: HP Original Ink
Ink Color: Black
Ink Type: Cartridge
[neo] -----
[neo] BUILD SUCCESS
[neo] -----
[neo] Total time: 1.816 s
[neo] Finished at: 2026-07-20T15:10:48+05:30
[neo] -----

```

Question 3

[pom.xml](#)

```
<project xmlns="http://maven.apache.org/POM/4.0.0"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
https://maven.apache.org/xsd/maven-4.0.0.xsd">
  <modelVersion>4.0.0</modelVersion>

  <groupId>com.workshop</groupId>
  <artifactId>university-list-injection</artifactId>
  <version>1.0.0</version>

  <properties>
    <maven.compiler.release>17</maven.compiler.release>
    <project.build.sourceEncoding>UTF-8</project.build.sourceEncoding>
  </properties>

  <dependencies>
    <dependency>
      <groupId>org.springframework</groupId>
      <artifactId>spring-context</artifactId>
      <version>6.1.14</version>
    </dependency>
  </dependencies>

  <build>
    <plugins>
      <plugin>
        <groupId>org.apache.maven.plugins</groupId>
        <artifactId>maven-compiler-plugin</artifactId>
        <version>3.13.0</version>
      </plugin>
      <plugin>
        <groupId>org.codehaus.mojo</groupId>
        <artifactId>exec-maven-plugin</artifactId>
        <version>3.5.0</version>
        <configuration>
          <mainClass>com.workshop.university.App</mainClass>
        </configuration>
      </plugin>
    </plugins>
  </build>
</project>
```

[applicationContext.xml](#)

```
<?xml version="1.0" encoding="UTF-8"?>
<beans xmlns="http://www.springframework.org/schema/beans"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="http://www.springframework.org/schema/beans
https://www.springframework.org/schema/beans/spring-beans.xsd">

  <bean id="university" class="com.workshop.university.model.University">
    <property name="universityName" value="Noida Institute of Engineering and
Technology" />
  </bean>
</beans>
```

```

        <property name="departments">
            <list>
                <value>Computer Science</value>
                <value>Information Technology</value>
                <value>Electronics and Communication</value>
                <value>Mechanical Engineering</value>
                <value>Civil Engineering</value>
            </list>
        </property>
    </bean>
</beans>

```

[University.java](#)

```

package com.workshop.university.model;

import java.util.List;

public class University {
    private String universityName;
    private List<String> departments;

    public String getUniversityName() {
        return universityName;
    }

    public void setUniversityName(String universityName) {
        this.universityName = universityName;
    }

    public List<String> getDepartments() {
        return departments;
    }

    public void setDepartments(List<String> departments) {
        this.departments = departments;
    }

    public void displayDepartments() {
        System.out.println("University Name: " + universityName);
        System.out.println("Departments:");

        for (String department : departments) {
            System.out.println(department);
        }
    }
}

```

[App.java](#)

```

package com.workshop.university;

import org.springframework.context.ApplicationContext;
import org.springframework.context.support.ClassPathXmlApplicationContext;

import com.workshop.university.model.University;

```

```

public class App {
    public static void main(String[] args) {
        ApplicationContext context = new
ClassPathXmlApplicationContext("applicationContext.xml");
        University university = context.getBean("university", University.class);
        university.displayDepartments();
    }
}

```

Output

```

[INFO] Scanning for projects...
[INFO] -----< com.workshop:university-list-injection >-----
[INFO] Building university-list-injection 1.0.0
[INFO] from pom.xml
[INFO] -----[ jar ]-----
[INFO] --- clean:3.2.0:clean (default-clean) @ university-list-injection ---
[INFO] Deleting E:\Programming\college_workshop\3rd_Year\SEM_V\Advance_Java_Workshop\day_7\Question_3\universityListInjection\target
[INFO] --- resources:3.4.0:resources (default-resources) @ university-list-injection ---
[INFO] Copying 1 resource from src/main/resources to target/classes
[INFO] --- compiler:3.11.0:compile (default-compile) @ university-list-injection ---
[INFO] Recompiling the module because of changed source code.
[INFO] Compiling 2 source files with javac [debug release 17] to target/classes
[INFO] --- exec:3.5.0:java (default-cli) @ university-list-injection ---
University Name: Noida Institute of Engineering and Technology
Departments:
Computer Science
Information Technology
Electronics and Communication
Mechanical Engineering
Civil Engineering
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 2.020 s
[INFO] Finished at: 2026-07-20T15:18:39+05:30
[INFO] -----

```

Question 4

[pom.xml](#)

```
<project xmlns="http://maven.apache.org/POM/4.0.0"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
https://maven.apache.org/xsd/maven-4.0.0.xsd">
  <modelVersion>4.0.0</modelVersion>

  <groupId>com.workshop</groupId>
  <artifactId>course-enrollment-system</artifactId>
  <version>1.0.0</version>

  <properties>
    <maven.compiler.release>17</maven.compiler.release>
    <project.build.sourceEncoding>UTF-8</project.build.sourceEncoding>
  </properties>

  <dependencies>
    <dependency>
      <groupId>org.springframework</groupId>
      <artifactId>spring-context</artifactId>
      <version>6.1.14</version>
    </dependency>
  </dependencies>

  <build>
    <plugins>
      <plugin>
        <groupId>org.apache.maven.plugins</groupId>
        <artifactId>maven-compiler-plugin</artifactId>
        <version>3.13.0</version>
      </plugin>
      <plugin>
        <groupId>org.codehaus.mojo</groupId>
        <artifactId>exec-maven-plugin</artifactId>
        <version>3.5.0</version>
        <configuration>
          <mainClass>com.workshop.course.App</mainClass>
        </configuration>
      </plugin>
    </plugins>
  </build>
</project>
```

[Course.java](#)

```
package com.workshop.course.model;

import org.springframework.stereotype.Component;

@Component
public class Course {
  private int courseId = 501;
  private String courseName = "Advanced Java";
}
```

```

        private String facultyName = "Dr. Sharma";

        public int getCourseId() {
            return courseId;
        }

        public String getCourseName() {
            return courseName;
        }

        public String getFacultyName() {
            return facultyName;
        }
    }
}

```

[Student.java](#)

```

package com.workshop.course.model;

import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.stereotype.Component;

@Component
public class Student {
    private int studentId = 101;
    private String studentName = "Aarav Sharma";

    @Autowired
    private Course course;

    public int getStudentId() {
        return studentId;
    }

    public String getStudentName() {
        return studentName;
    }

    public Course getCourse() {
        return course;
    }

    public void displayStudentDetails() {
        System.out.println("Student Details");
        System.out.println("Student ID: " + studentId);
        System.out.println("Student Name: " + studentName);
        System.out.println("Course ID: " + course.getCourseId());
        System.out.println("Course Name: " + course.getCourseName());
        System.out.println("Faculty Name: " + course.getFacultyName());
    }
}

```

[AppConfig.java](#)

```

package com.workshop.course.config;

import org.springframework.context.annotation.ComponentScan;
import org.springframework.context.annotation.Configuration;

```

```
@Configuration
@ComponentScan(basePackages = "com.workshop.course")
public class AppConfig {
}
```

App.java

```
package com.workshop.course;

import org.springframework.context.annotation.AnnotationConfigApplicationContext;

import com.workshop.course.config.AppConfig;
import com.workshop.course.model.Student;

public class App {
    public static void main(String[] args) {
        try (AnnotationConfigApplicationContext context = new
AnnotationConfigApplicationContext(AppConfig.class)) {
            Student student = context.getBean(Student.class);
            student.displayStudentDetails();
        }
    }
}
```

Output

```
[INFO] Scanning for projects...
[INFO] -----< com.workshop:course-enrollment-system >-----
[INFO] Building course-enrollment-system 1.0.0
[INFO] from pom.xml
[INFO] -----[ jar ]-----
[INFO] --- clean:3.2.0:clean (default-clean) @ course-enrollment-system ---
[INFO] Deleting E:\Programming\college_workshop\3rd_Year\SEM_V\Advance_Java_Workshop\day_7\Question_4\CourseEnrollmentSystem\target
[INFO] --- resources:3.4.0:resources (default-resources) @ course-enrollment-system ---
[INFO] skip non existing resourceDirectory E:\Programming\college_workshop\3rd_Year\SEM_V\Advance_Java_Workshop\day_7\Question_4\CourseEnrollmentSystem\src\main\resources
[INFO] --- compiler:3.11.0:compile (default-compile) @ course-enrollment-system ---
[INFO] Recompiling the module because of changed source code.
[INFO] Compiling 4 source files with javac [debug release 17] to target\classes
[INFO] --- exec:3.5.0:java (default-cli) @ course-enrollment-system ---
Student Details
Student ID: 101
Student Name: Aarav Sharma
Course ID: 501
Course Name: Advanced Java
Faculty Name: Dr. Sharma
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 2.113 s
[INFO] Finished at: 2026-07-20T15:24:31+05:30
[INFO] -----
```